Secondary Examination, 2015

Mathematics

Time- $3\frac{1}{4}$ **Hours**

M.M- 70

GENERAL INSTRUCTIONS TO THE EXAMINEES:

1) There are in all seven questions in this question paper.
2) All the questions are compulsory.
3) In the beginning of each, the number of parts to be attempted has been clearly mentioned.
4) Marks allotted to the questions are indicated against them.
5) Start from the first question and proceed to the last one.
6) Do not waste your time over a question you cannot solve

Q.1(a) Evaluate the value is $\frac{50x^2-98y^2}{10x-14y}$.

Sol.

Simplify we have the expression is $\frac{50x^2-98y^2}{10x-14y}$.

We take 2 common are nemoninator and denominator.

$$\frac{2\left(25x^2-49y^2\right)}{2\left(5x-7y\right)}$$

$$\frac{\left(25x^2-49y^2\right)}{\left(5x-7y\right)}$$

We know that,

$$\left(a\right)^2-\left(b\right)^2=\left(a+b\right)\left(a-b\right)$$

$$\frac{\left(5x+7y\right)\left(5x-7y\right)}{\left(5x-7y\right)}=\left(5x+7y\right)$$

Hence, the solution is $\left(5x+7y\right)$.

(b). Find the value of $\frac{\sin 20^0}{\cos 70^0}$.

Sol

Simplify the expression is $\frac{\sin 20^0}{\cos 70^0}$.

We know that $\sin(90-\theta)=\cos\theta$.

$=\frac{\sin\left(90^0-70^0\right)}{\cos 70^0}$

$=\frac{\cos 70^0}{\cos 70^0}=1$

Hence, the solution is 1.

(c). Determine the 1 to 10 mean of positive odd number .

Sol.

Simplify the expression.

We know that 1 to 10 positive odd number is 1, 3, 5, 7.

Mean = $\frac{\textit{total}\text{ number of sum}}{\textit{number}\text{ of term}}$

$=\frac{1+3+5+7}{4}$

$=\frac{16}{4}=4$

Hence, we get the value of 4.

(d). which is the following is the discount in the act 80 G.

(a) National saving certificate. (b) National security fund.

(c) General provident fund. (d) Future life premium.

Sol.

80 G discount act following is national security fund.

Hence, the solution is (national security fund).

(e). In figure, AB is diameter of circle and O is the center of circle. If $\angle COB = 50^0$ then, find the value of <CAB.

(i) 20^0 (ii) 25^0 (iii) 40^0 (iv) 60^0

Sol.

We have the angle is $\angle COB = 50^0$ and obtain the value is <CAB.

Angle $\angle COB = 50^0$, then angle $\angle AOC = 130^0$

We know that the angle of straight line is

Then, angle B 180^0

$\angle AOC$ is $180^0 - 50^0 = 130^0$

Therefore we have the <OAC and <OCA are similar triangle.

Since, the angle $\angle CAB = 25^0$

$180^0 - 130^0 = 50^0$

$$\frac{50^0}{2} = 25^0$$

Angle <OAC and <OCA are 25^0.

Since, the angle $\angle CAB = 25^0$

(f). if $x + \frac{1}{x} = 7$, then find the value of $x^2 + \frac{1}{x^2}$

Sol.

Simplify the expression we have $x + \frac{1}{x} = 7$.

We know that $(a+b)^2 = a^2 + b^2 + 2ab$

Then,

$$x+\frac{1}{x}=7$$

$$\left(x+\frac{1}{x}\right)^2=7^2$$

$$x^2+\frac{1}{x^2}+2x\cdot\frac{1}{x}=49$$

$$x^2+\frac{1}{x^2}+2=49$$

$$x^2+\frac{1}{x^2}=49-2$$

$$x^2+\frac{1}{x^2}=47$$

Hence, we get the solution is $x^2+\frac{1}{x^2}=47$.

2. **All question are compulsory.**

(a) Determine the quadratic equation is $x^2+kx+3=0$ find the value of k if the value of x is 1.

Sol.

Simplify the expression now, we have quadratic equation $x^2+kx+3=0$.

We know that general quadratic equation is $ax^2+bx+c=0$.

Putting the value of $x=1$ in given equation $x^2+kx+3=0$

$$(1)^2+k(1)+3=0$$
$$k+4=0$$
$$k=-4$$

Hence, the value of $k=-4$.

(b). If the $\sin\theta=\frac{3}{5}$, then find the value of $\tan\theta$.

Sol.

We have the value of $\sin\theta=\frac{3}{5}$.

Now, $\sin\theta=\frac{3}{5}$ taking the value of $\tan\theta$ in theorem of Pythagoras.

We know that Pythagoras theorem $base^2 + perpendicular^2 = hypotenous^2$

$$5^2 - 3^2 = x^2$$
$$25 - 9 = x^2$$
$$x^2 = 16$$
$$x = 4$$

Let x= perpendicular

$$5^2 - 3^2 = x^2$$
$$25 - 9 = x^2$$
$$x^2 = 16$$
$$x = 4$$

Therefore,

$$\tan\theta = \frac{3}{4}$$

Hence, the value of $\tan\theta = \frac{3}{4}$.

(C). Devas buy a digital camera which is the cost of $6,000$ Rs. And after he is selling of 7% profit. What is the selling price of digital camera.

Sol.

We have the digital cost of camera is $6,000$.

We know that $x\% = \frac{x}{100}$ then, similarly $7\% = \frac{7}{100}$.

Simplify further,

$$6,000 \times \frac{7}{100} = 420$$
$$6,000 + 420 = 6420$$

Hence, the selling prince of digital camera is 6420 Rs.

(d). If 27, 23, x-4, x+4, 15, 3 and 7 of arithmetic value is 15. when, find the value of x.

Sol.

Simplify the arithmetic value is.

We know that the formula of mean below is.

$$mean = \frac{total \text{ number of sum}}{number \text{ of term}}$$

$$15 = \frac{27 + 23 + x - 4 + x + 4 + 15 + 3 + 7}{7}$$

$105 = 75 + 2x$

$2x = 105 - 75$

$2x = 30$

$x = 15$

Hence, the value of mean is 15.

3**. Q Solute the all question.**

(a). Find the value of point the interception by a line $3x + 4y = 12$.

Sol.

Simplify the expression.

$3x + 4y = 12$

Let,

$x = 0$

Putting $x = 0$, solve the x-intercept in equation $3x + 4y = 12$.

$3(0) + 4y = 12$

$= (0,\ 3)$

Let, $y = 0$, solve y- intercept putting the value of y in equation $3x + 4y = 12$.

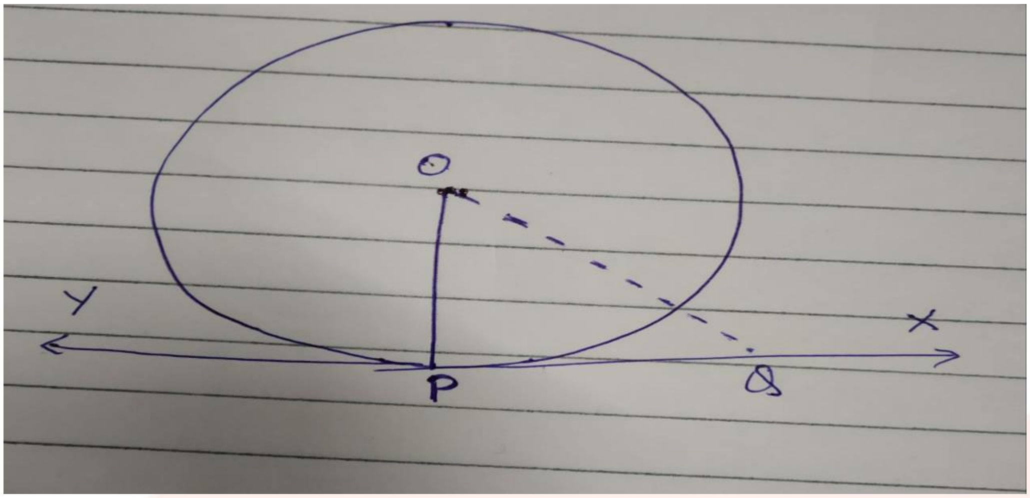
$3x + 4(0) = 12$

$(4,\ 0)$

Hence, the intercept point $(0,\ 3)$ and $(4,\ 0)$.

(b). The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Sol.



We have a circle with Centre O and tangent XY to the circle at point P. we required to prove that OP is perpendicular to XY.

We take point Q on XY other than P and join OQ.

The point Q must lie outside the circle, and note that if Q lies inside the circle. XY will become a secant and not a tangent to the circle. Since, OQ is longer than the radius OP of the circle. That is,

$$OQ > OP.$$

Therefore this happen for every point on the line XY except the point P, OP is the shortest of all the distance of the point O to the points of XY . then is OP is perpendicular to XY.

(c). find the value of. $\tan\left(\frac{13\pi}{3}\right)$

Sol .

Simplify the expression we have the value of $\tan\left(\frac{13\pi}{3}\right)$.

Subtract (2π) in value of $\tan\left(\frac{13\pi}{3}\right)$ and simplify further.

$$x = \frac{13\pi}{3} - 2\pi$$
$$x = \frac{7\pi}{3}$$

We get the resulting angle of $\left(\frac{7\pi}{3}\right)$ is positive less than (2π).

Again subtract (2π) in value of $\left(\frac{7\pi}{3}\right)$.

$$x = \frac{7\pi}{3} - 2\pi$$
$$x = \frac{\pi}{3}$$
$$\tan\frac{\pi}{3} = \tan 60^0$$
$$\tan 60^0 = \sqrt{3}$$
$$= \sqrt{3}$$

Hence, we get the value of $\sqrt{3}$.

(d). find the curve surface area two cylinder whose radius is same and their height in the ratio 3:2.

Sol. simplify the expression.

We know that the cylinder surface area is $2\pi rh$.

Simplify the further ratio of surface area of cylinder.

$$\frac{2\pi rh_1}{2\pi rh_2} =$$
$$\frac{2\pi r \cdot 3}{2\pi r \cdot 2} = \frac{3}{2}$$
$$= \frac{3}{2}$$

Hence, we get the value of surface area ratio $3:2$.

4. Solute the all question.

(a). find the line diagonal which is perpendicular on $3x-4y+8=0$

Sol.

Simplify the expression.

$$3x-4y+8=0$$
$$-4y=3x+8$$
$$-y=\frac{3x+8}{4}$$
$$-y=-\left(\frac{3x}{4}-2\right)$$

Since, we know that when two line are perpendicular. The product of their slopes is -1.

$$\frac{3}{4}\times m=-1$$
$$m=\frac{-4}{3}$$

Hence, the line diagonal value is $\frac{-4}{3}$.

(b) Evaluate the value of $\frac{2\tan 15^0}{1+\tan^2 15^0}$.

Sol.

Simplify the expression.

We know that,

$$\tan\theta=\frac{\sin\theta}{\cos\theta}$$

Similarly, $\tan 15^0 = \dfrac{\sin 15^0}{\cos 15^0}$

We know that,

$$\tan 15^0 = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

Substituting the value of $\tan 15^0$ in given expression.

$$\frac{2\tan 15^0}{1+\tan^2 15^0}$$

$$= \frac{2\cdot\left(\dfrac{\sqrt{3}-1}{\sqrt{3}+1}\right)}{1+\left(\dfrac{\sqrt{3}-1}{\sqrt{3}+1}\right)^2}$$

Simplify further,

$$= \frac{\dfrac{\left(4-2\sqrt{3}\right)}{8}}{\left(4+2\sqrt{3}\right)}$$

$$= \frac{(4)^2-\left(2\sqrt{3}\right)^2}{8} = \frac{16-12}{8}$$

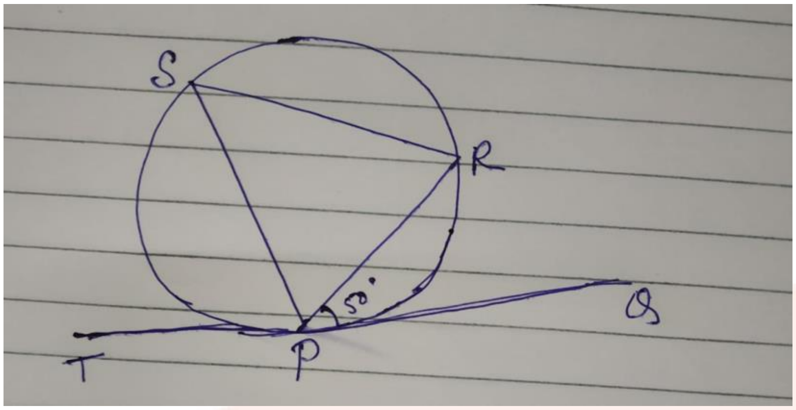
$$= \frac{4}{8} = \frac{1}{2}$$

Hence, the solution value is $\dfrac{1}{2}$.

(c) figure in. O is the centre of circle and TPQ is a tangent line. If $\angle RPQ = 50^0$, so find the angle $\angle PSR$

.

Sol.



$\angle RPQ = 50^0$

We know that

$\angle OPQ = 90^0$

Then,

$OPR = 90^0 - 50^0 = 40^0$
$\angle OPR = \angle ORP$
$= 40^0$

Therefore,

$\angle OPR + \angle POR + \angle ORP = 180^0$
$40^0 + 40^0 + \angle POR = 180^0$
$\angle POR = 100^0$
$\angle PSR = \frac{1}{2}\angle POR$
$\angle PSR = 50^0$

(d). find the slant height of cone. whose given cone height and diameter of 12cm and 18cm.

Sol.

Simplify the slant height of cone.

We have the cone diameter 18cm and height of cone 12cm.

We know that the radius of half diameter.

Therefore the radius get is 9cm.

Using the formula of slant height is.

$L=\sqrt{r^2+h^2}$

Substituting the value of r and h in above formula.

$L=\sqrt{9^2+12^2}$
$L=\sqrt{81+144}$
$L=\sqrt{225}$
$L=15$

Hence, the solution is slant height of cone 15.

5. Q All question are solute.

(a). prove that: $\cos 4A=1-8\sin^2 A+8\sin^4 A$

Sol.

Simplify the expression we have this equation $\cos 4A=1-8\sin^2 A+8\sin^4 A$.

L.H.S

$$\begin{aligned}
\cos 4A &= \cos 2\cdot 2A \\
&= \cos^2 2A-\sin^2 2A \\
&=1-\sin^2 2A-\sin^2 2A \\
&=1-2\sin^2 2A \\
&=1-2\left(2\sin A\cdot\cos A\right)^2 \\
&=1-2\left(4\sin^2 A\cdot\cos^2 A\right) \\
&=1-8\sin^2 A\cdot\cos^2 A \\
&=1-8\sin^2 A\left(1-\sin^2 A\right) \\
&=1-8\sin^2 A+8\sin^4 A=R.H.S
\end{aligned}$$

Hence, $R.H.S=L.HS$ proved

(b). prove that $\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}$. If the original point on the line which is the intersection of the end axis.

Sol.

Prove that expression.

Equation of line AB in intercept form.

$\frac{x}{a}+\frac{y}{b}=1$, $\perp$ distance form is (0,0) in P.

Therefore $\perp$ distance $=\left|\frac{Ax_1+By_1+c}{\sqrt{A^2+B^2}}\right|$

C is the constant value.

$$p=\left|\frac{\frac{1}{a}\times 0+\frac{1}{b}\times 0}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}\right|$$

$$p=\left|\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}\right|$$

$$\frac{1}{p}=\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$$

Squaring both side.

$$\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}$$

Hence, the expression proved.

(c). find the median from the following frequency distribution.

Class interval	$0-12$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency s	6	9	12	8	15

Sol.

We know that $x_1=\frac{0+12}{2}=6$ similarly $x_2, x_{3---}x_5$

And, $\sum f_1x_1=f_1\times x_2$

C.I	f_1	x_1	f_1x_1
0-12	6	6	36

10-20	9	15	135
20-30	12	25	300
30-40	8	35	280
40-50	15	45	675
	$\sum f_1 = 50$		$\sum f_1 x_1 = 1426$

$mean = \frac{1426}{50} = 28 \cdot 52$

Hence, the value of mean $28 \cdot 52$.

(d) In year 2013-14, the annual income of x is 62,0,000 (excluding house rent). He deposits its 8000 per month in his usual future fund account. He doposits 80,000 in his PDF account. He deposits 80,000 in his PDF account. Calculate the income tax payable by X while the maximum limit of savings of saving is one lakh. Income tax rates are as follows:

Apart from this 3% education levy is levied on the income tax payable.

	INCOME	INCOMETAX
(i)	UP to Rs 2,00,000	0%s
(ii)	Rs 2,00,001 to Rs5,00,000	Rs 2,00,000 increase to income of 10%
(iii)	Rs 5,00,001 to Rs 10,00,000	Rs 30,000 +Rs 5,00,000 increase to income 20%

Sol.

X of yearly income = $Rs\ 6,20,000$.

Future saving income = per monthly $Rs\ 8{,}000 \times 12$

Rs 96,000

PPF saving account income = Rs 80,000 +96,000+1,00,000=Rs 27,6000.

Total saving income = $Rs\ 6,20,000 - 2,76000 = Rs\ 3{,}44000$

Calculate the income tax

$3,44,000 \times 10\%$

$= 3,44,000 \times \frac{10}{100} = 34,000$

Total income tax $= 34,000$

$= 34,000 \times 3\%$

$= 34,000 \times \frac{3}{100}$

$= 1020$

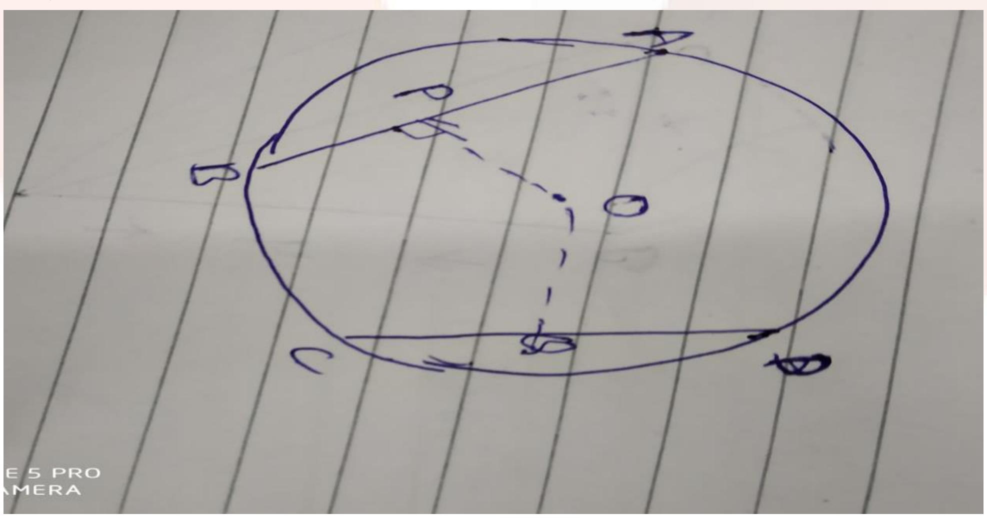
Total pay income tax $= 34,000 + 1020 = 35,020$

Hence, the value of total pay income tax $35,020$

6 All question are solute.

(a) A chords equilibrant from the center of a circle are equal in length.

Sol.



We have to drawn two chords AB & CD which are equilibrant
Given AB & CD equilibrant $OP = OQ$
To prove = $AB = CD$
Contraction –join OA & OC.
Proof-

$$\text{In } \begin{array}{l} \Delta OAP, \ \&\Delta \text{OCQ} \\ \angle\text{OPA}=\angle\text{OQC}=90 \\ \text{OP=OQ} \\ \text{\&OA=OC} \\ \Delta\text{OAP} \cong \Delta OCQ = R.H.S \end{array}$$

Congruence rule.

$AB = CD$

Hence, proved it.

(b) Find the value of P. if $Ax + By = C$ and $x\cos\alpha + y\sin\beta = p$ are denote parallel line.

Sol.

We know that, line equation

$$y = mx + c$$

$$y = \frac{-A}{B}x + \frac{C}{B}$$

$$y = \frac{-x\,\text{Cos}\,\alpha}{\text{Sin}\,\alpha} + \frac{P}{\text{Sin}\,\alpha}$$

$$\frac{A}{B} = \frac{\text{Cos}\,\alpha}{\text{Sin}\,\alpha},$$

$$\frac{C}{B} = \frac{P}{\text{Sin}\,\alpha}$$

Further, comparing this,

$$P = \frac{C\,\text{Sin}\,A}{B}$$

Hence, the value is $P = \frac{C\,\text{Sin}\,A}{B}$.

(c) find the radius. Whose given the cone slant height h is 13cm and the hole surface area is 90π sm.

Sol.

Simplify the expression .

We know that the cone hole surface area is.

$A = \pi r(l + r)$

$90\pi = \pi r(13 + r)$

$90 = r(13 + r)$

$90 = 13r + r^2$

$r^2 + 13 - 90 = 0$

Using the formula is quadratic equation.

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-13 \pm \sqrt{13^2 - 4 \cdot 1 \cdot 90}}{2 \cdot 1}$$

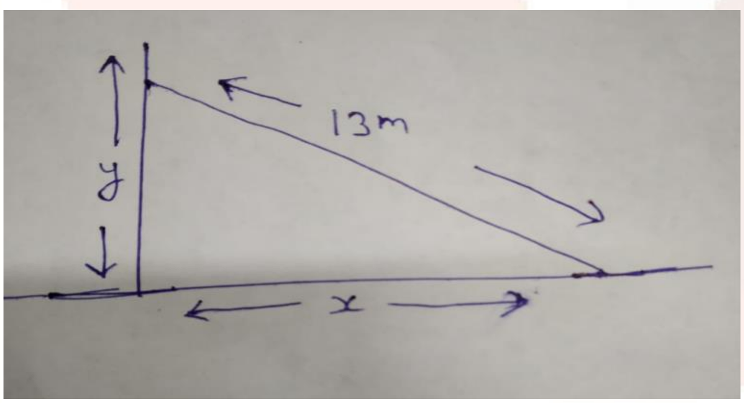
$$r = \frac{-13 \pm \sqrt{169 - 360}}{2}$$

$$r = \frac{-13 \pm \sqrt{-191}}{2}$$

Hence the value of r $r = \dfrac{-13 \pm \sqrt{-191}}{2}$.

(d). The angle of elevation of a tower to the point O is 15^0. on travelling 80m,towards the tower the angle will increase to 30^0 .So find the height of the tower .

Sol.



Let AB the height of tower and DC is 80

We take the total distance BD is $x + 80$

Take the angle 30^0

$\tan 30^0 = \dfrac{h}{x}$

And taking the angle 15^0

$\tan 15^0 = \dfrac{h}{80+x}$

We know that $\tan 15^0 = \dfrac{\sqrt{3}-1}{\sqrt{3+1}} = \left(2-\sqrt{3}\right)$

$$\left(2-\sqrt{3}\right) = \frac{h}{80+x}$$
$$\left(2-\sqrt{3}\right)80 + x = h$$
$$160 + 2x - 80\sqrt{3} - \sqrt{3}x = h$$

Substituting the value of h.

$$160 + 2x - 80\sqrt{3} - \sqrt{3}x = \frac{x}{\sqrt{3}}$$
$$160\sqrt{3} + 2\sqrt{3}x - 240 - 30x = x$$
$$160\sqrt{3} + 2\sqrt{3}x - 240 = 4x$$
$$4x - 2\sqrt{3}x = 160\sqrt{3} - 240$$
$$x = \frac{160\sqrt{3} - 240}{4x - 2\sqrt{3}x} = \frac{80\sqrt{3} - 120}{2-\sqrt{3}}$$

Hence, the height of tower $\dfrac{80\sqrt{3}-120}{2-\sqrt{3}}$.

(D).find the P. If $Ax + By = C$ and $x\cos\alpha + y\sin\alpha = p$ are fixed in parallel line.

Sol.

Simplify the expression we have the value is $x\cos\alpha + y\sin\alpha = p$

$x\cos\alpha + y\sin\alpha = p$

Squaring both side above this equation.

$$\left(x\cos\alpha + y\sin\alpha\right)^2 = p^2$$
$$x^2\cos^2\alpha + y^2\sin^2\alpha = p^2$$

7. All question are solute.

(a). Evaluate $\left(\frac{x-1}{x+1}\right)^4 - 13\left(\frac{x-1}{x+1}\right)^2 + 36 = 0$.

Sol.

Simplify the expression.

$$\left(\frac{x-1}{x+1}\right)^4 - 13\left(\frac{x-1}{x+1}\right)^2 + 36 = 0$$

Let, $\left(\frac{x-1}{x+1}\right)^2 = x$

We get this equation $x^2 - 13x + 36 = 0$.

Factorize $x^2 - 13x + 36 = 0$ this equation.

$$x^2 - 13x + 36 = 0$$
$$x^2 - 9x - 4x + 36 = 0$$
$$x(x-9) - 4(x-9) = 0$$
$$= (x-9)(x-4)$$
$$x - 9 = 0$$
$$x = 9$$
$$x - 4 = 0$$
$$x = 4$$

Hence, the solution is.

(b) The sum of two digits number is 12.If the digits are interchanged, it is found that the resulting number is greater than 18 by the original number. What is the two digit number.

Sol.

Simplify the expression.

Then, the interchange the digit the new number will be 10x+y. It is found that the resulting new number is greater than the original number by 18.

$x+y=12......(i)$

So, $$\begin{aligned}&10x+y=10y+x+18\\&9x-9y=18\\&9(x-y)=2\\&x-y=2.......(ii)\end{aligned}$$

Simplify the equation 1 and equation 2 obtain the value of x and y.

$x+y=12.$

$x-y=2$

$x=7$

Putting the value of x in equation 1.

$7+y=12$

$y=5$

since, we get the value is 75.

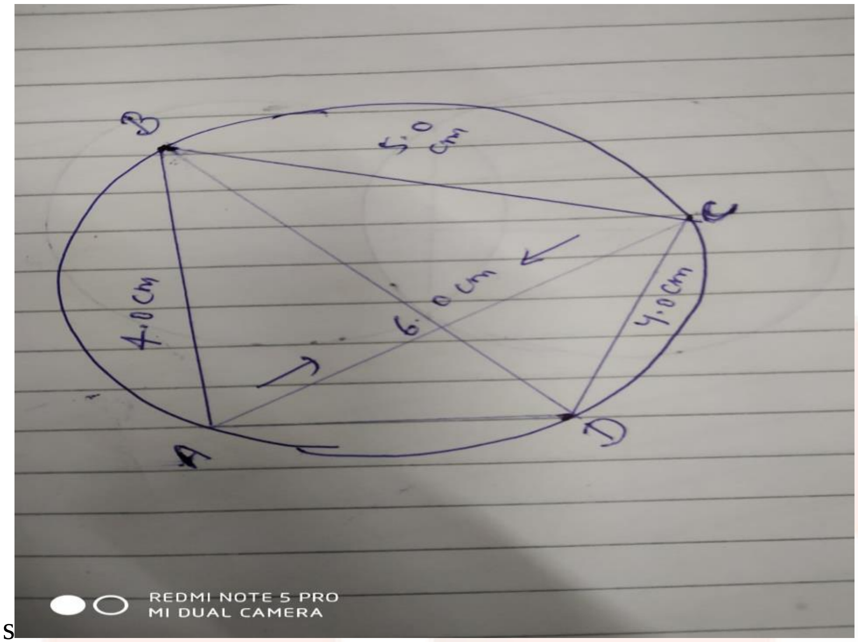
Hence, the solution is 75.

b) Draw a cyclic quadrilateral ABCD. Given the $AB=4.0\,\text{cm}$, $AB=5.0\,\text{cm}$, $AC=6.0\,\text{cm}$, and $CD=4.0\,\text{cm}$.

sol.

we know that a cyclic quadrilateral ABCD. A quadrilateral whose verticles all lie on a single circle. This circle is called the circumcirle or circumscribed circle, and the vertices and said to be concylic.

We have $AB=4.0\,\text{cm}$, $AB=5.0\,\text{cm}$, $AC=6.0\,\text{cm}$, and $CD=4.0$.



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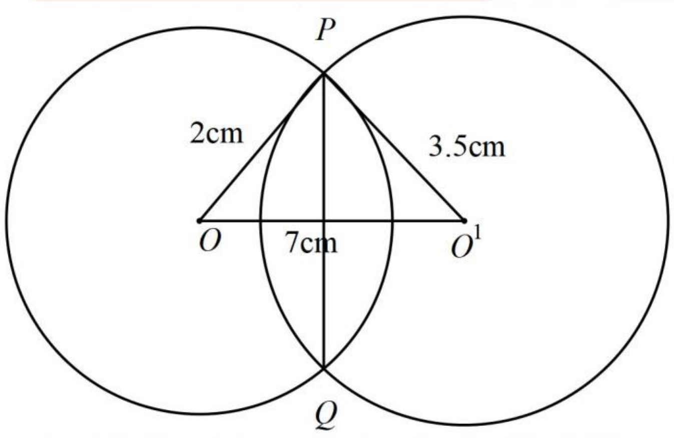
Or

Q. Two circle with centre O and O' of radius 2cm and 3.5cm. respectively intersect at two point p and Q such that OP and O'P are tangent to circle . whose distance are two circle in 7cm, find the common parallel line.

Sol.



Here, OP and OP'are two tangent drawn at point.

$\angle OPO' = 90^0$

Tangent at any point-of circle is perpendicular to radius through the point of contct.

Joint

$\left(OO^{'}\right)$ and $\left(PN\right)$. In right angled

$\angle ONP,$

$2^2 + x^2 = \left(3\cdot 5\right)^2 + \left(7 - x\right)^2$

$4 + x^2 = 12\cdot 25 + 49 + x^2 - 14x$

$x = 4$

Putting the value x in equation $\left(7 - x\right)$.

$\left(7 - x\right) = \left(7 - 4\right) = 3$

We get ON is 4 and NO' 3.

Simplify further,

$\angle ONP$

$OP^2 = ON^2 + NP^2$

$4^2 - 3^2 = PN^2$

$PN^2 = 4^2 - 3^2 = 16 - 9 = 7$

$PN = \sqrt{7}$

Therefore, we get $PN = \sqrt{7}$. So PQ $PQ = 2\sqrt{7}$.

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